

IdeateCOMM UI/UX Design

2024

Cheat sheet 25th September 2024

eig.



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Introduction to UI/UX Design

User Interface

UI, or User Interface, refers to the space and as its name suggests, the interface between the user/ human and the machine, where interactions occur. In digital products, this can refer to anything from its visuals, audio, interactions such as buttons, text fields, images, sliders, etc. The primary purpose of the UI is to enable users to navigate, control, and accomplish tasks within a system.

The 4 guiding principles which we will cover include:

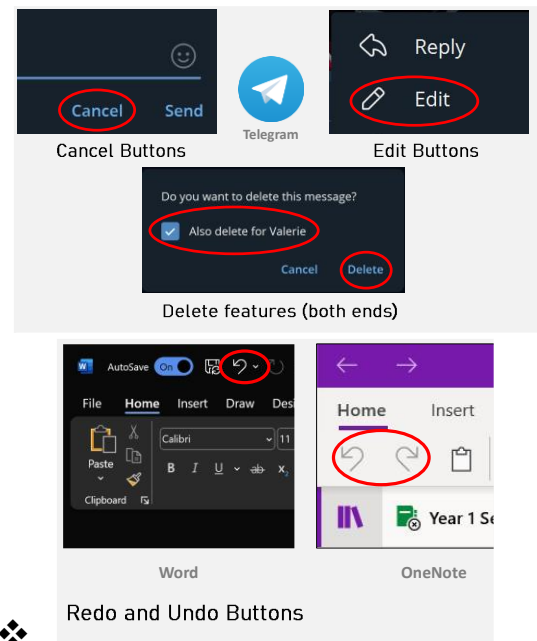
1. Users in control (User-Centric Designs)
2. Comfortable Interactions
3. Cognitive Ease
4. Consistent Interface

I. *Users in control/ User-Centric Designs*

Any good UI should be user-centered in its design. It should aim to cater to the user's preferences and needs, while striking a balance in aesthetics and functionality, empowering users to have control and autonomy during their interactions with the interface. A few tips that can help you with this include:

❖ *Allowing for do-overs*

Users make mistakes, and giving them the ability to undo, cancel, or edit these actions helps them **recover easily from those errors**. This limits the possibilities of irreversible consequences, and **reduces frustration**, in turn giving them **more confidence** to explore and engage with the UI. This also offers the user **flexibility in their choices**, allowing for **better decision-making**.

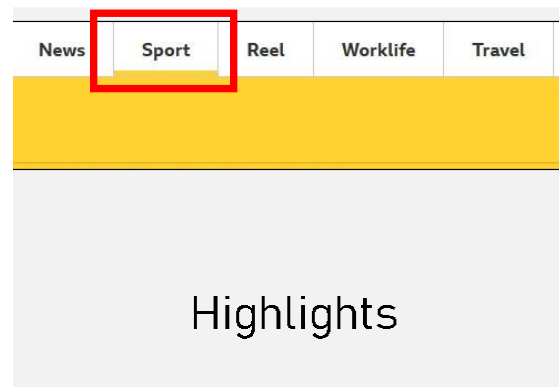


Providing visual cues in UI design is important as it serves to **improve overall usability and user experience (UX)** by guiding users and making the interface more intuitive. A few examples include page titles, and highlights in a webpage.

Page titles help **orientate** users of where they are within the app or website, helping them understand the current context and navigate confidently, **reducing confusion**. Highlights or other visual cues help **guide users' attention** to important elements, helping them quickly identify key actions, information, or areas that require focus. This **reduces cognitive load** and helps users **prioritize tasks**. All these helps to **improve accessibility** for easier navigation and **enhance its usability**.

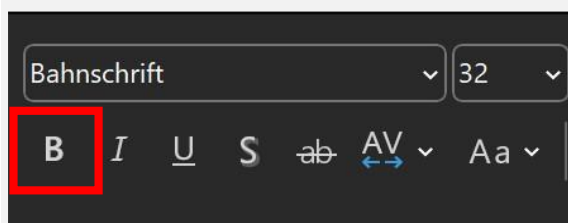


Page Titles



❖ Predictability

Predictability in UI design is crucial in reducing confusion and error-making. When buttons work as expected/ intended, over time, this builds mental models and supports user learning and memory of navigation of said UI. This reduces errors, builds user confidence in your product and UI, and thus creates trust between the user and the system. By creating this trust and reducing confusion (thus preventing frustration), this directly contributes to user satisfaction and the overall user experience.



Buttons should perform as intended/ expected

i.e. "B" icons **bold texts**, **not italicise** / underline them

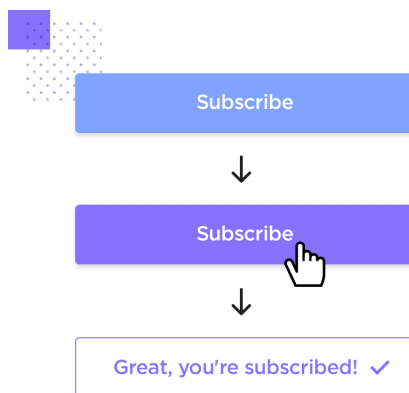
❖ *Visibility of system status*

Visibility of system status is a fundamental principle in UI design. It keeps the user updated on what the system is doing and **reduces uncertainty**. It serves as feedback for the user, so it either **reassures users** that the system is processing their request, or **guides users on further actions to be taken** to rectify any issue that arises. This **reduces frustration**, and **builds trust in the system's reliability**, promoting **transparency and inspiring confidence** in the product.



❖ *Feedback after action*

Implementing feedback after actions **ensures clarity**, **builds user confidence**, and makes the interface feel more **responsive and intuitive**. For example, when a button is pressed, in a lot of UIs nowadays, the button either changes color, or enlarges accordingly, or something happens, to convey feedback. This provides reassurance to the user that the input has been recognized and reduces confusion and uncertainty. Over time, this contributes to building user confidence.

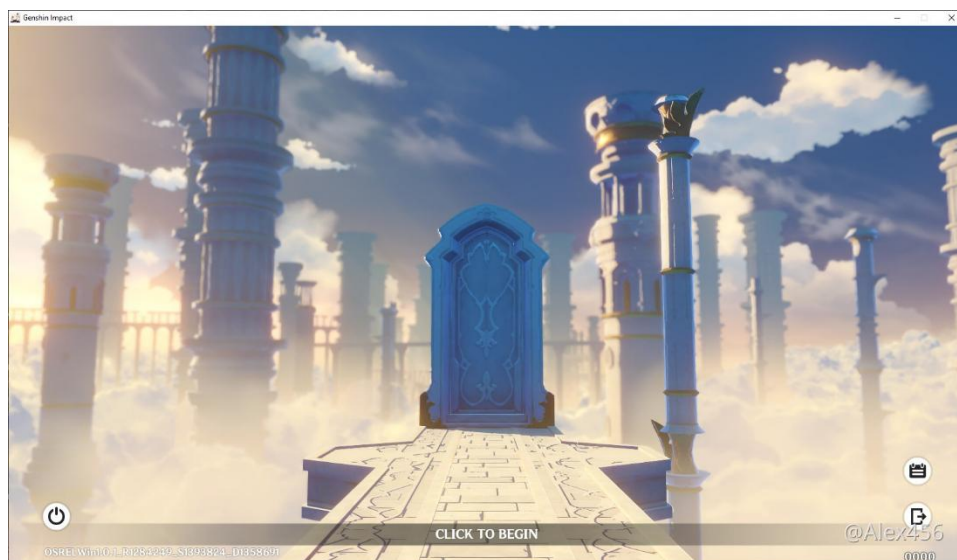


Change of color
and message as a
way of interaction
by the buttons

❖ *Accommodate to everyone*

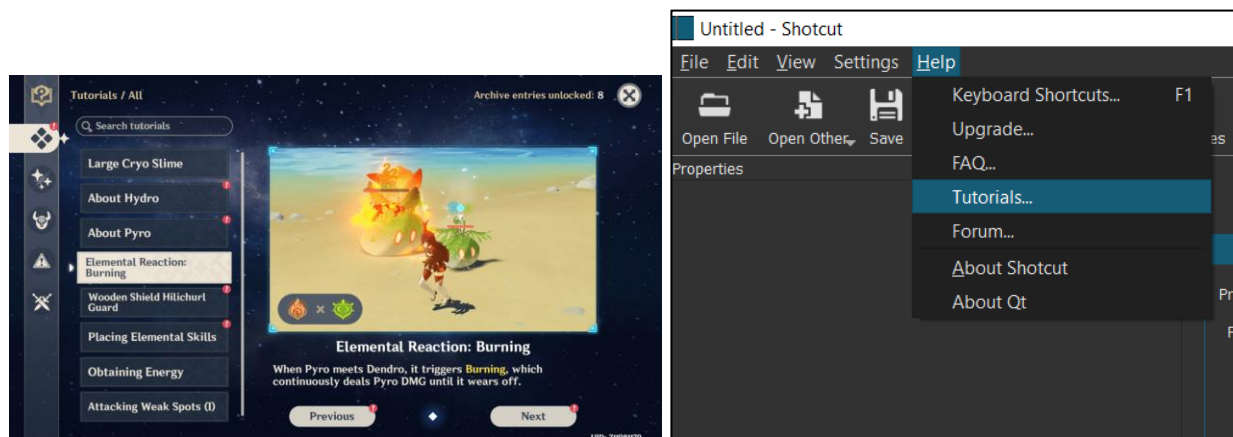
This **ensures inclusivity, promotes accessibility, and makes it usable** for people with varying needs and abilities. For example, tactile feedback and voice/ speak to chat ensures that visually impaired people too, can use the product you are trying to sell. Likewise, ensuring buttons change colors upon pressing, and tactile feedback, accommodates to people with hearing impairment.

This also means accommodating to varying levels of skills. There are several ways around this, such as making instructions simple, but clear and effective. Example, in the big hit game Genshin Impact, this was how they made instructions simple, and easy to follow:

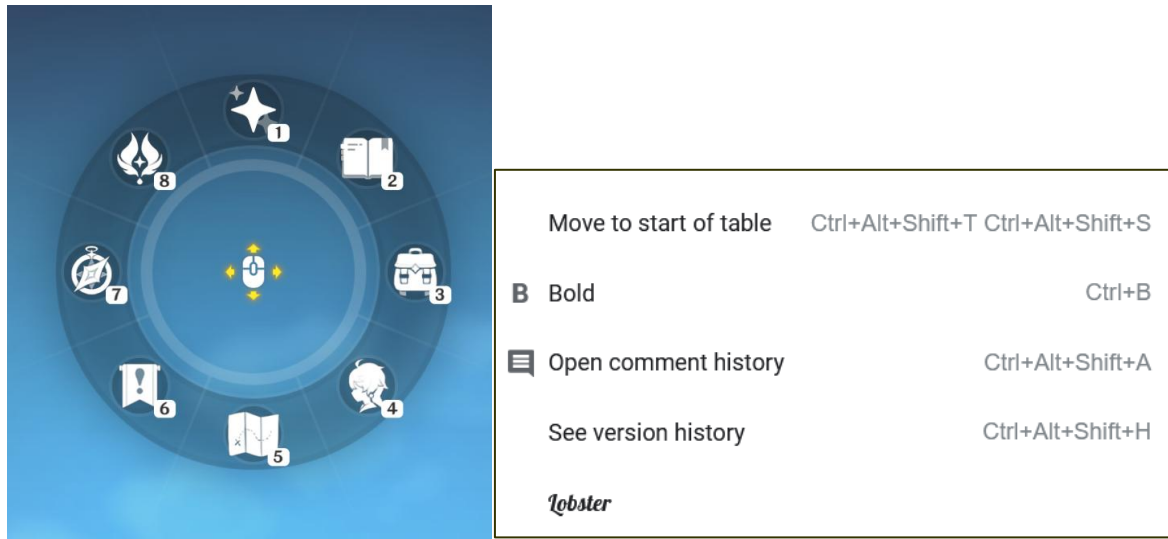


"Click to begin"

Other methods include introducing tutorials and help for beginners,



Or shortcuts for experienced users.



II. *Comfortable Interactions*

A good UI, since its user-centered in its design, should consider the many interactions the user would have with the system, and the user interface that links both together. Thus, it would be ideal for interactions to be made as comfortable as possible, as this **reduces frustration, ensures cognitive ease, promotes user satisfaction and enhances the overall user experience**. 3 core guiding principles that help with this include **simplicity, visual hierarchy, and designing a UI to be fault tolerant**.

➤ *Simplicity*

Simplicity is an important aspect in UI/UX Design, as a simple interface serves to create an **intuitive and user-friendly experience**. By keeping things simple, it **minimizes the mental load and toll** that the user would take on as they attempt to interact and navigate the UI. This thus also **promotes accessibility** and helps guide users to where you wish to direct them to (look up designing with persuasive patterns, and choice architecture) / where they wish to be directed to, resulting in **faster decision-making** as well. A bad example of this will include:

Form 4506-T
(September 2018)
Department of the Treasury
Internal Revenue Service

Request for Transcript of Tax Return

▶ Do not sign this form unless all applicable lines have been completed.
▶ Request may be rejected if the form is incomplete or illegible.
▶ For more information about Form 4506-T, visit www.irs.gov/form4506t.

OMB No. 1545-1872

Tip. Use Form 4506-T to order a transcript or other return information free of charge. See the product list below. You can quickly request transcripts by using our automated self-help service tools. Please visit us at irs.gov and click on "Get a Tax Transcript..." under "Tools" or call 1-800-908-9946. If you need a copy of your return, use Form 4506, Request for Copy of Tax Return. There is a fee to get a copy of your return.

1a Name shown on tax return. If a joint return, enter the name shown first.	1b First social security number on tax return, individual taxpayer identification number, or employer identification number (see instructions)
2a If a joint return, enter spouse's name shown on tax return.	2b Second social security number or individual taxpayer identification number if joint tax return
3 Current name	
4 Previous address shown on the last return filed if different from line 3 (see instructions)	
5a If the transcript or tax information is to be mailed to a third party (such as a mortgage company), enter the third party's name, address, and telephone number.	
5b Customer file number (if applicable) (see instructions)	

Who even reads this?? How is this relevant for the average user?

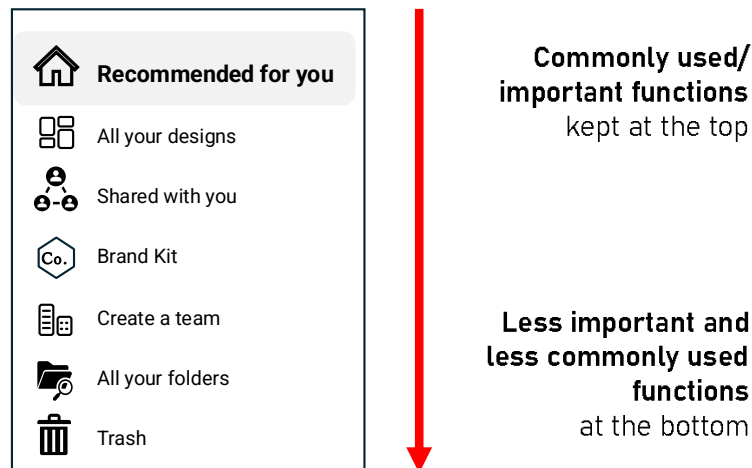
Not user-friendly

Designed to assist admins, not general user

Blah Blah
Blah...

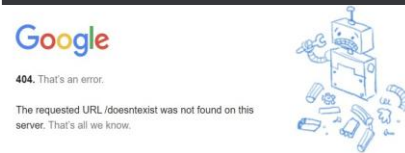
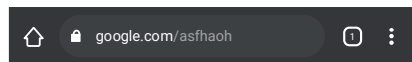
➤ Visual Hierarchy

This is regarding the arrangement of elements within the design of the UI. By considering what are some of the common features that most users will want, we can then prioritize said elements accordingly. This is important as it **directs users' attention** and helps them focus on important elements first, **enhancing readability** while indirectly **conveying the relative importance** of certain information.

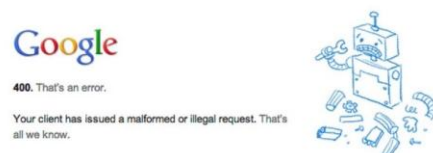


➤ Fault-Tolerant Designs

Fault tolerance in UI design refers to the system's ability to handle user errors without causing frustration. This is crucial in reducing frustration, minimize error impact by offering safeguards, and helps to accommodate users of varying needs and skill levels. The aspect of “allowing for do-overs” also falls under here, but this specifically refers to the broader category. A good fault-tolerant design will prompt users to know how to correctly rectify the error, or at least offer the option on how to proceed thereafter. A bad example however will be if the follow-ups on the matter are vague or misleading.

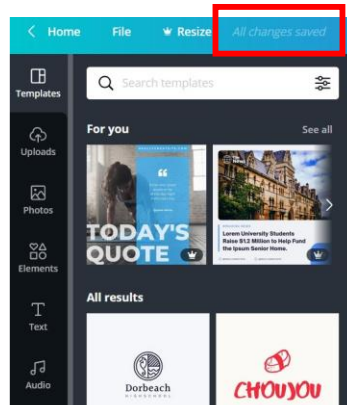


Error message is clear,
Guides users on next action



Error message is vague

A good example of fault-tolerant designs would be the autosave features commonly observed in Microsoft products, as well as Canva.



Work is
saved
constantly,
reducing
chances of
work loss

III. *Cognitive Ease*

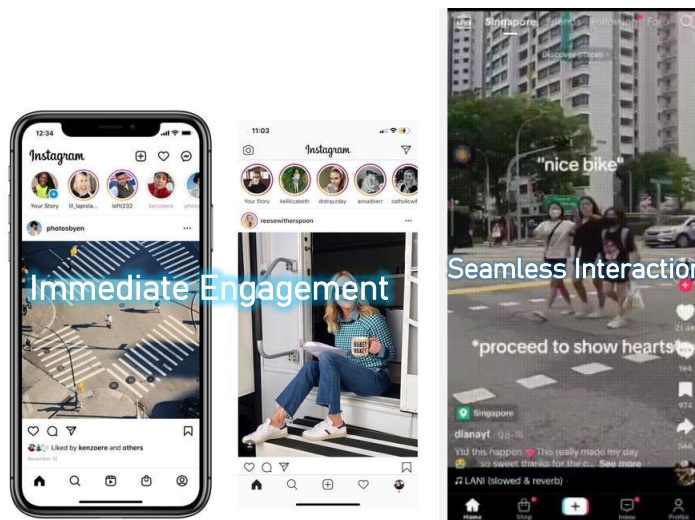
It is ideal to keep the mental effort required for users to navigate and interact with your UI to be as low as possible. When users feel cognitive ease with your UI, it leads to **increased user satisfaction without users feeling mentally drained**.

➤ *Frictionless Design: Endless Scroll & Infinite Content*

Endless scrolling and content being consistently fed to the user, results in **no ends/endpoints**. Clever visual tricks also hint at more content awaiting just off-screen, cueing you to keep scrolling for more novel, emotional, and informative videos. Coupled with a low effort (frictionless) design (1 swipe just for the next video), it makes it **easy for users to continue and lose track of time in the process**. This frictionless interaction **contributes to addictive behavior** (refer to below, habit formation) and makes it harder for the consumer to disengage.

➤ *Frictionless Design: Immediate Engagement*

By reducing any barriers between the user and their desired outcome, it **reduces the hassle** and **helps to feed instant gratification and dopamine release**. (Refer to below) For example, TikTok begins playing vibrant videos immediately on open rather than showing a boring home feed. You don't have to click on anything to be entertained right away (Immediate Engagement). **Instead of choice paralysis on where to start, amusement is delivered instantly**, all in all further contributing to increased user satisfaction and a better UX design.



➤ *Frictionless Design: 3-Click Rule*

Users should be able to find any information on websites, or applications within just 3 clicks. This ties into the **usability and easy navigation** of the UI itself, contributing to a positive user experience. By avoiding overly complicated paths, with many steps, or questions along the way, users will not be turned away from using the app or website due to the hassle they will need to go through with.

➤ *Chunking Pieces of information*

Chunking pieces of info

+65	9868	5985
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Easier to process, understand, & remember ✓

Easier to check for errors ✓

In cognitive psychology, by chunking pieces of information, it **enhances readability** by presenting information in “**bite-sized**” pieces.

➤ *Clarity: Grid System*

Clarity in how information is presented is important in ensuring information can be **presented neatly** for the users, so they can **process, understand & remember**. It also serves to help **avoid information overload** for the user. A bad example of this would be the old “The New York Times” newspaper.



One way we can work around this would be the usage of a grid system. It is a layout tool that organizes content into a **structured layout**, ensuring that elements are **aligned and spaced consistently** throughout the interface.



Top horizontal grid shows the most important information

The Guardian

Articles of similar importance aligned vertically



The Guardian

IV. Consistent Interface

A consistent interface provides visual clarity, and **primarily ties into helping with cognitive ease in navigation of UI**. An inconsistent interface will throw users off or result in them being more mentally fatigued while navigating your UI, affecting overall user experience. A bad example of this includes:



A few key principles that help with this, includes:

- Visual Clarity & Readability
 - ❖ White Space
 - ❖ Color Theory
 - ❖ Iconography
 - ❖ Avoiding Jargon & Technical Terms
 - ❖ Typography
- Functional Consistency

Let's take a look:

➤ *Visual Clarity & Readability*

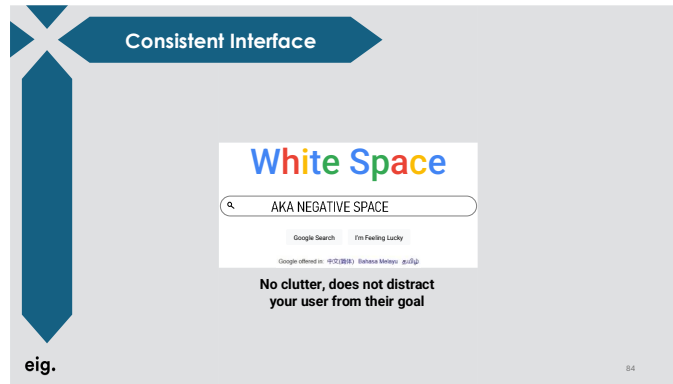
Visual Clarity refers to the measure of how effectively visual design prioritizes and conveys information. If it's insufficient, users can be quickly overwhelmed by the UI they need to navigate and seem more difficult to them than it actually is.

Readability is about the reader – the ease with which a reader can successfully decipher, process, and make meaning of the text read. Typography ties heavily into this, ranging from font choice, sizes, and spacing. Good readability means users get to read and understand content—efficiently—and helps influence their decision-making process in a good way.

❖ *Clarity: Whitespace*

White space is one of the 13 basic design principles and refers to any blank or empty space surrounding all the other elements in any design. It is the space between text, images, buttons and other objects that a user can see on a page or a screen. It is also known as negative space, and despite its name, it is not always “white” in color. It just refers to any empty space devoid of any other visual elements, such as circles, squares, or icons.

The main purpose of white space often times is to separate, emphasize and help accentuate (make more prominent/ “more obvious”) other design elements. It is also to help ensure visual harmony and legibility/ readability. There are 2 main forms of whitespace.



As observed, whitespace can also help to prevent clutter and ensure user is not distracted from their goal.

Micro Whitespace



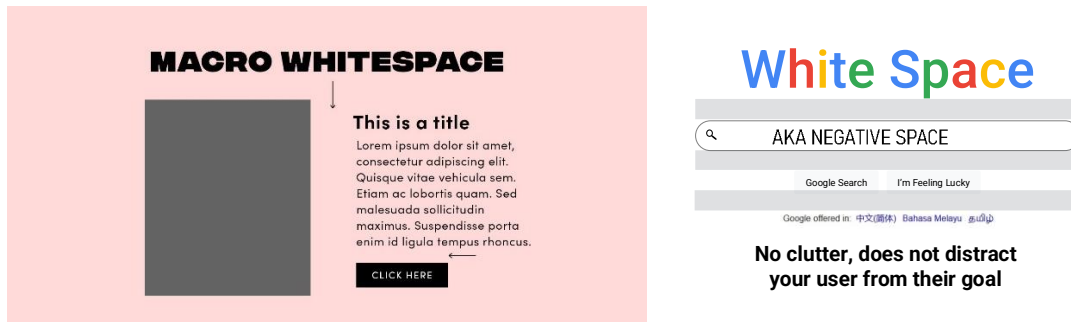
Micro Whitespace refers to the small instances of empty spaces, such as in between each character, lines of text, and images. It is often subtle and used to ensure text and other visual elements are legible, and don't look too cluttered.



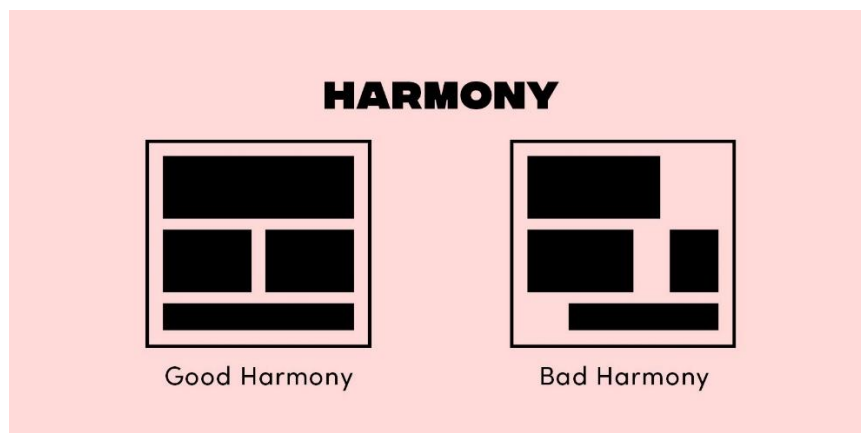
Letters that are too closely packed or too widely spaced apart can be difficult to read. While there are no hard and fast rules to follow when it comes to letter and line spacing, if it takes effort to read, the spacing likely needs to be adjusted.

Macro Whitespace

Macro whitespace refers to larger areas of empty space in a design. This is the more common application of whitespace in design. Examples of macro whitespace include the margins on the left and right of websites or slides design. It helps add to clarity and contribute to visual harmony, and often times helps to accentuate or highlight important visual elements you would like the user to notice.



Visual Harmony is important as it creates an effect of unity, and completeness. As observed below, macro whitespace can also help to structure, organize and cluster content.



❖ *Clarity: Color Theory*

Color Theory primarily contributes to visual harmony and clarity, via the idea of color harmony. A pleasing arrangement of colors to the eye helps to engage the user and offers balance in UI design. It can also help to contrast and accentuate visual elements. The idea of harmony is important as it serves as a balance: If something is bland and boring, users will not engage with your UI. If something is overly stimulating, the user will feel overwhelmed and confused. The human brain rejects what it cannot organize (under-stimulation; bland and boring), or what it cannot understand (over-stimulation; overwhelmed and confused). Color Harmony, that is built upon Color Theory, helps to deliver visual interest, and a sense of structure. There are various categories of color wheel, namely built using primary, secondary and tertiary colors.



Primary Colors



Secondary Colors



Tertiary Colors

Primary Colors: *Red, yellow and blue*

In traditional color theory (used in paint and pigments), primary colors are the 3 pigment colors that cannot be mixed or formed by any combination of other colors. All other colors are derived from these 3 hues.

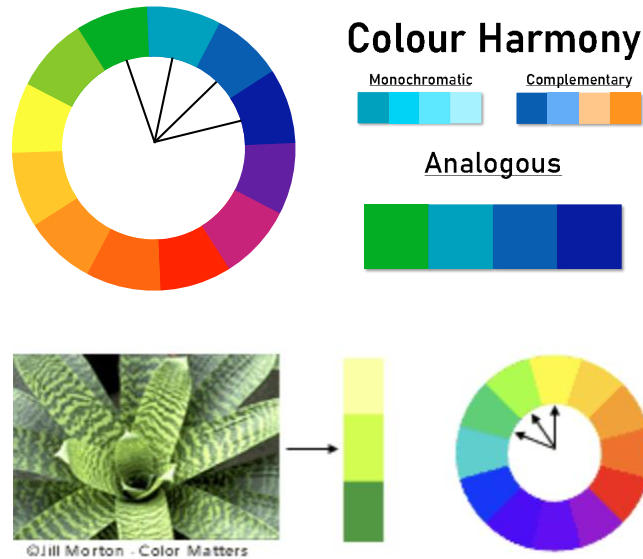
Secondary Colors: *Green, orange and purple*

These are the colors formed by mixing the primary colors.

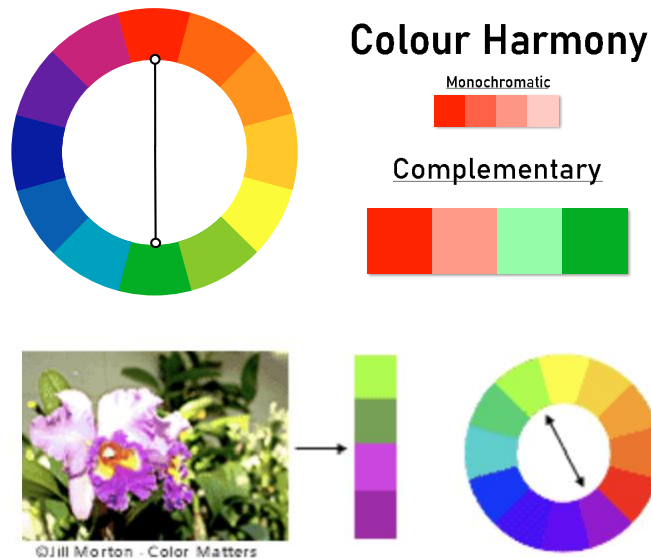
Tertiary Colors: *Yellow-orange, red-orange, red-purple, blue-purple, blue-green & yellow-green*

These are the colors formed by mixing a primary and a secondary color.

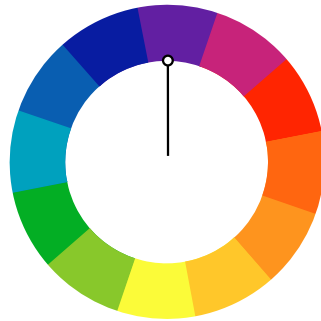
The key ideas behind Color Theory are **analogous colors**, **complementary colors** and **monochromatic colors**.



2 examples of **analogous colors**: It represents any 3-4 colors (ideally 3) that are side by side on a 12-part color wheel, where usually 1 color predominates. (As observed in the first picture above, green is a conflicting dominant color as compared to dark blue.)



Complementary colors are any two colors which are directly opposite each other, such as red and green, and red-purple and yellow-green. It serves to offer maximum contrast and stability. As observed above, there are several variations of yellow-green in the leaves and several variations of red-purple in the orchid.



Colour Harmony

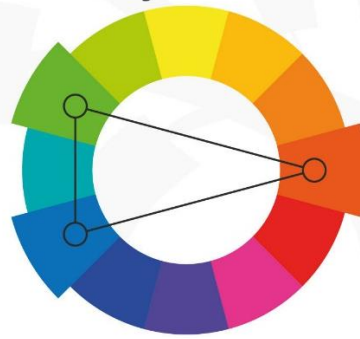
Monochromatic



Monochromatic helps to ensure consistency in UI design, and convey simplicity and elegance, while simplifying visual hierarchy.

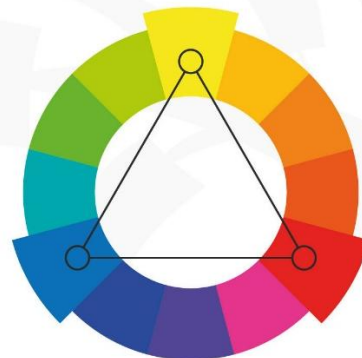
Other implementations of color theory also include **Split-Complementary (Compound Harmony)**, **Triadic**, **Tetradic**, **Square**, or sticking to the basic **Warm & Cool colors**.

Split-Complementary



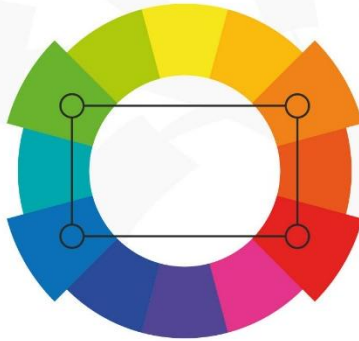
Compound Harmony utilizes the addition of colors from either side of your complementary color pair to soften the contrast.

Triadic



Triadic takes 3 equally distant colors on the color wheel (i.e., 120° apart: e.g., red/blue/yellow) helps to maintain harmony and high contrast.

Tetradic



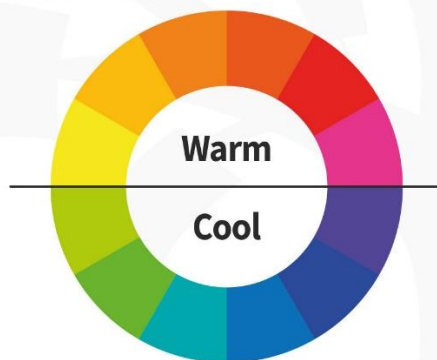
Tetradic colors take four colors that are two sets of complementary pairs (e.g., orange/yellow/blue/violet) and choose one dominant color. It allows for rich, contrasting and lesser seen designs, but it is difficult to balance well. It is necessary to achieve a balance between warm and cool colors.

Square



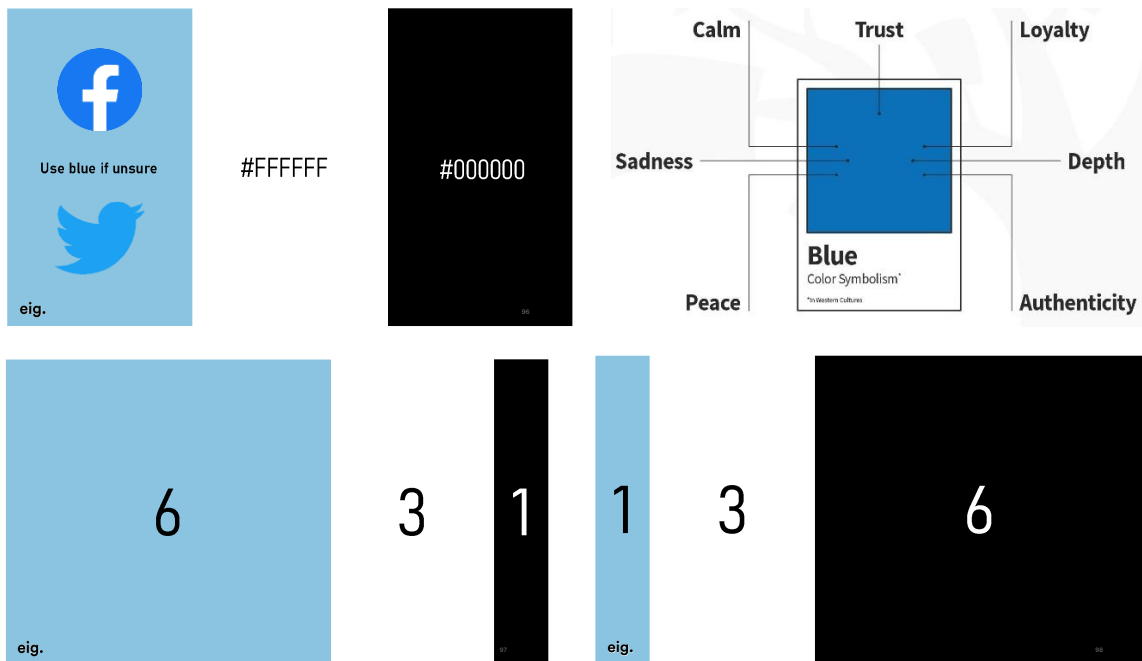
Square colors are a variant of tetradic, with four colors evenly spaced on the color wheel (i.e., 90° apart). Unlike tetradic, square schemes can work well if you use all four colors evenly.

Warm & Cool Colors

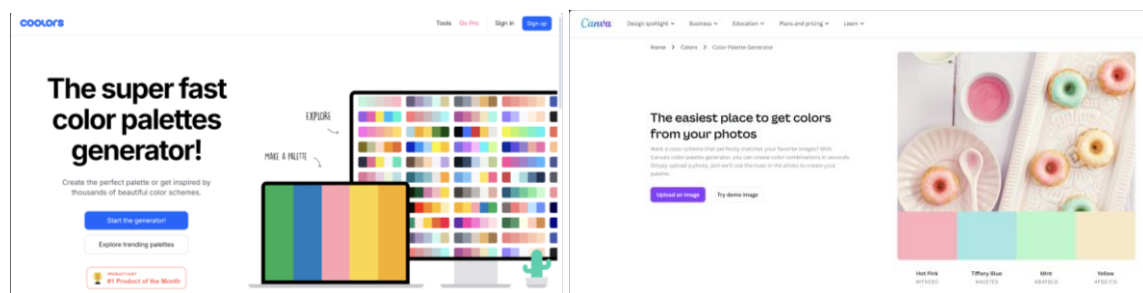


Warm and cool colors are easy to achieve color harmony since they share similar hues. They can also evoke certain emotions, or psychological responses. Warm colors are typically associated with energy, passion, warmth, excitement. Cool colors are typically associated with calmness, stability, trust and relaxation. **It is important that your colors reflect your design's goal and how the color temperature reflects your message.**

Another 3 tips: If you have trouble balancing, try to stick to 3 primary or secondary colors that can help in contrast (**Compound Harmony**). And also, **use blue color if unsure. Color ratio can be 6 : 3 : 1.**



A few recommended websites if you need help to get started on this:
Coolers & Canva, Interaction Design Foundation



❖ *Clarity: Iconography*

Iconography often uses icons, and sometimes symbols, to **convey and visually represent** actions, features, or content in an interface. They act as universal **visual cues** that assist in helping users **quickly navigate and understand the UI** without reliance on text descriptions. By instant recognition of common actions, it makes the UI **intuitive** and helps to **reduce cognitive load**.

Recognizable icons



Hang up



Camera **off**



Settings



Home



Mic



Alarm/Notifs

Visual Representation of UI

❖ *Readability: Avoiding Jargon & Technical Terms*

Using simple terms as compared to jargon (special words or expressions used by a profession or group that are difficult for others to understand) or technical terms, helps to ensure a wider audience can understand the UI. It also helps to ensure clarity in the messages, and that users can quickly navigate the UI without confusion, resulting in cognitive ease.

Example, *ceteris paribus* is used commonly in economics, but you won't say that to an engineering student who might not have had a background in economics who would not understand what it means.

Avoid Jargon

~~*Ceteris paribus*~~----- "All other things being equal"

~~*Primus inter pares*~~----- First amongst equals

~~*Carte Blanche*~~----- Freedom to act as one wishes

❖ Readability: Typography

Typography

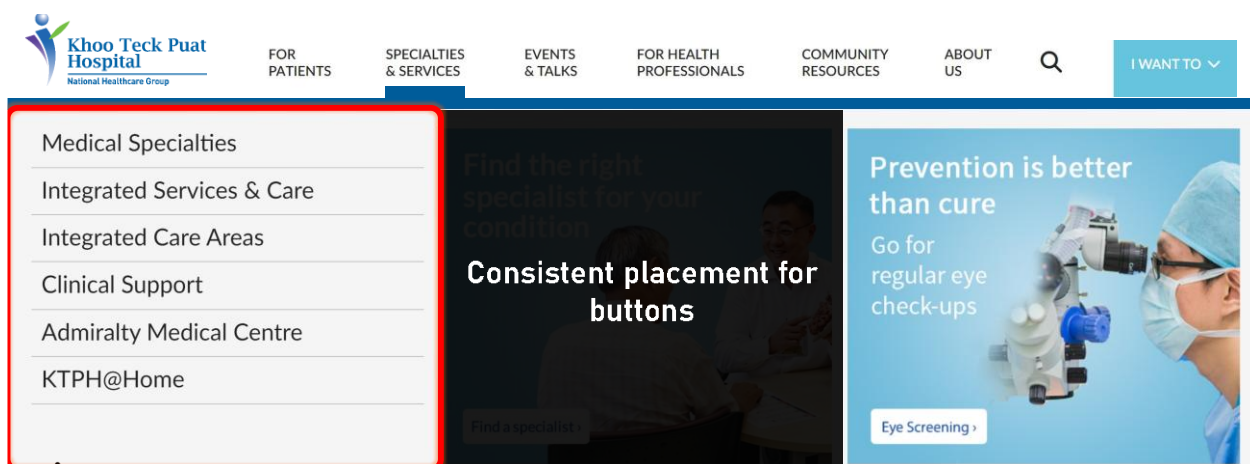
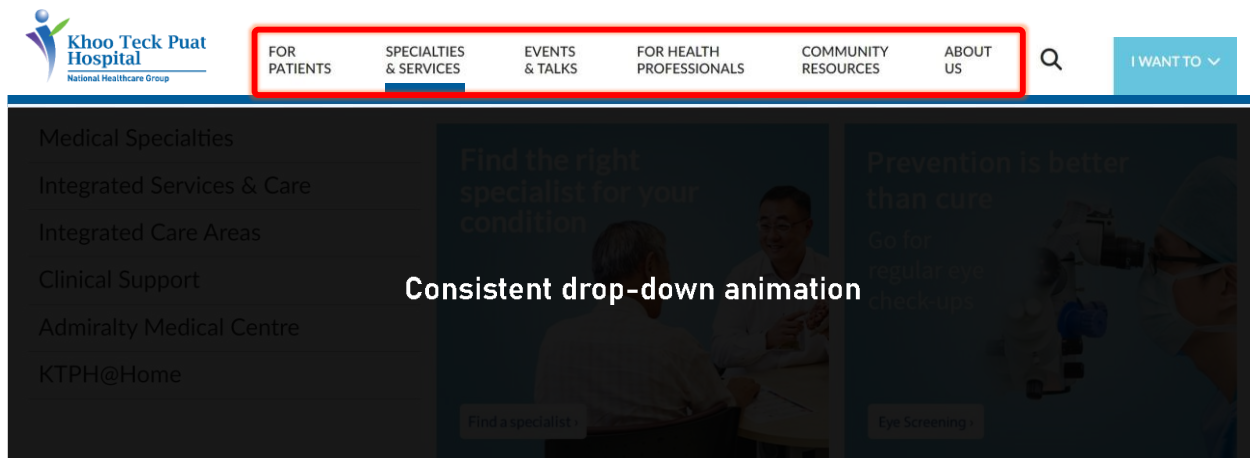
* Comic Sans MS is a HORRIBLE font choice

Typography

* Avoid Cursive Fonts!

Typography is an important aspect of the overall user interface and experience. Generally, users will prefer something they can read off easily, which promotes cognitive ease. A few tips we can give for this, includes highlighting, font colour choice, font sizes, as well as font choice. Try to avoid cursive fonts, and limit it to 3 or 4 different fonts in total.

➤ Functional Consistency



Try not to change common interfaces to avoid alienating users.

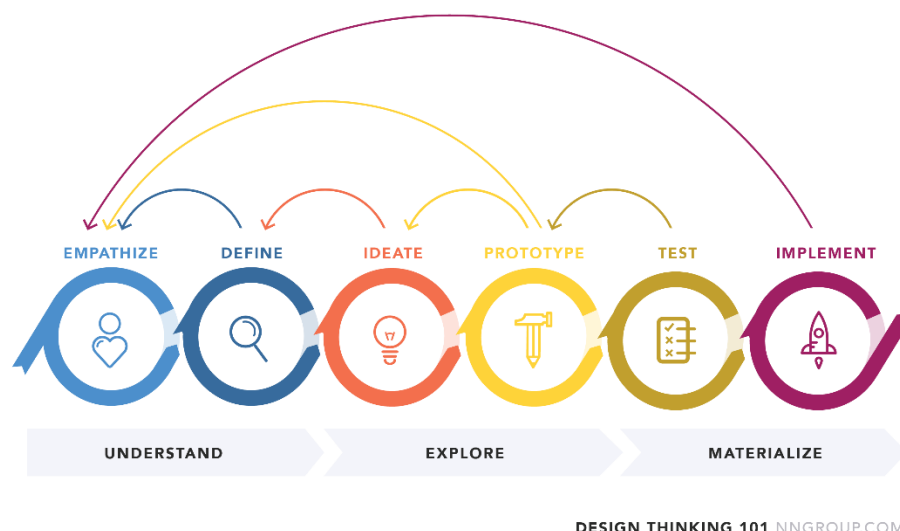
General Rule of Thumb:

If changing it makes it **more confusing**, do not change it.

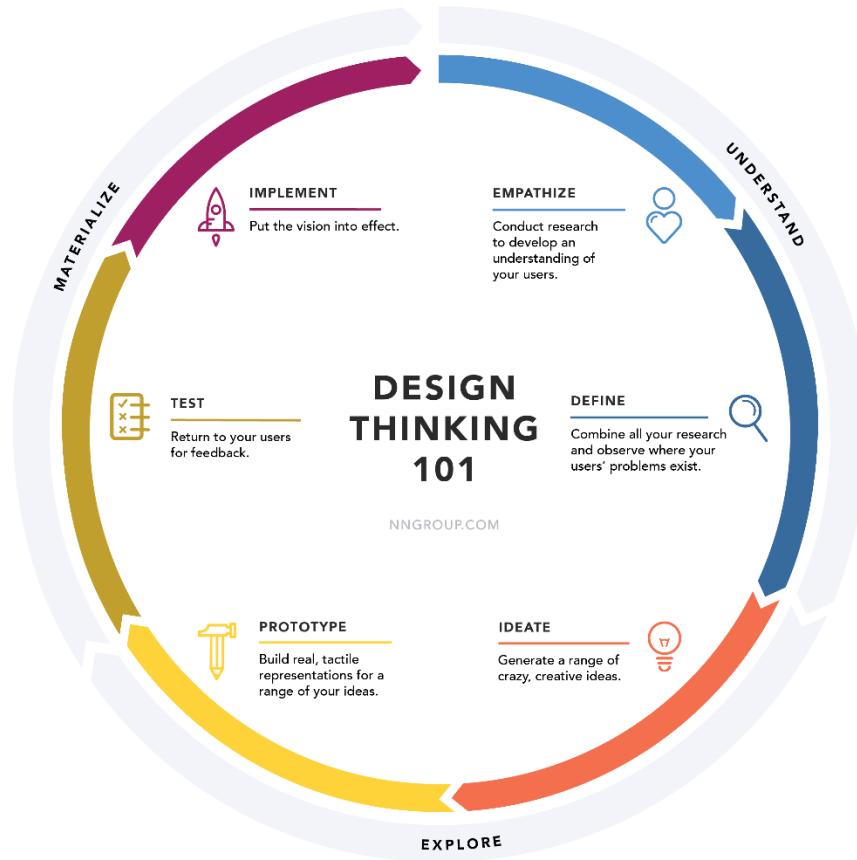
User Experience (UX)

User Experience is a broad term referring to the overall experience the user feels when navigating your product, system, or user-interface. It includes from the big things, such as the downloading of your application, the start-up screen, to even the small things like micro white spacing, or the ease of navigation. In this cheat sheet, we aim to equip you with the basics of what will be required for the frameworks of how to navigate and engineer the overall user experience for your users. A few things we will look at include an overall design thinking process framework, the honeycomb framework, as well as the laws of UI/UX design, and a few effects that result from UI/UX design.

Design Thinking Process



The design thinking ideology takes on a hands-on, user-centric approach to problem solving. The design thinking process refers to the framework in which this ideology is applied, and encompasses an overall flow of understanding the user, exploration of ideas, and materializing said idea.



i. *Empathize*

It is important to understand **who** your target audience is, and **what** they want, or **need**. It is recommended to conduct research in order to gain a deeper understanding about what your users do, say, think, and feel. Examples of guiding questions you can use, when you wish for them to navigate your website or application, such as “what motivates or discourages users?” or “where do they experience frustration?” The goal is to gather enough observations that you can truly begin to empathize with your users and their perspectives.

ii. *Define*

Collate your research and **observe** where your users’ problems exist. While pinpointing your users’ needs, begin to highlight opportunities for innovation.

iii. Ideate

Brainstorm a range of creative ideas that address the unmet user needs or wants identified in the earlier phases. Do not limit yourself to “unrealistic ideas” and try to come up with as many ideas as you can: The goal is to later see what works, what doesn’t work, and what can be combined together to make it come together and work well.

iv. Prototyping

Build a real, tactile, working prototype or representation for a subset of your ideas. The goal is to understand what components of your ideas work, and which do not. Weigh the impact vs. feasibility of your ideas through feedback on your prototypes.

v. Testing

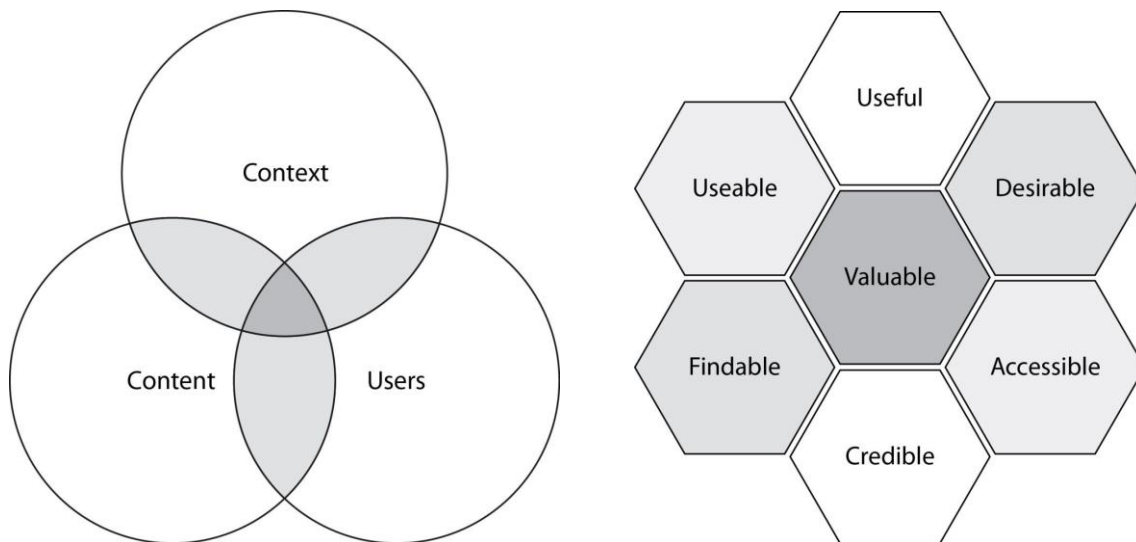
Return to your users for feedback. Put your prototype in front of real customers and verify that it achieves your goals. Consider their feedback on areas to be improved on, and what has been done well, and integrate it into your solutions accordingly. Also, ask yourself if the solution has addressed the original unmet needs.

vi. Implementation

Put the vision into effect. Ensure that your solution materialized and is put into prototyping and action. This is the most important part of design thinking, but it is not the end. After implementation, continue to return back to testing and user feedback, and continue to improve on the said solution.

Peter Morville's Honeycomb Design

Peter Morville is a designer and information architect who has been working in this field since 1994. He has held positions at top companies like Google and Gopher. While working as an IA, Peter began using a diagram of three circles that he felt best represented the connection between business goals and context, user needs and behavior, and content. It was then later further refined into the honeycomb design.



At the core of the honeycomb design, all the values tie down to how valuable the product is.

➤ *Usable*

The system or user interface that is delivered needs to be simple and easy to use. The design of the system should be familiar and easy to understand. The learning curve a user must go through should be minimal and painless as possible.

➤ *Useful*

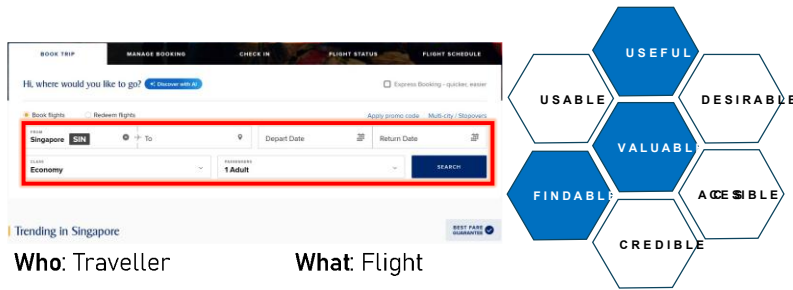
The product you are designing needs to be useful and fill a user's need. If the product has no real purpose, users do not have a real reason to pivot over into using your product.

➤ *Desirable*

The visual aesthetics of the UI/ product need to be attractive/ aesthetically pleasing and easy to navigate. Design should be minimal and to the point.

➤ *Findable*

Information, especially crucial information, needs to be findable and easy to navigate. If the user has problems, they should be able to quickly find a solution with minimal barriers.

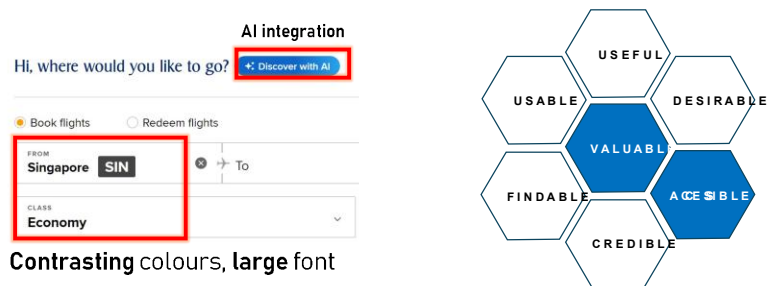


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➤ Accessible

The product or services should be designed so that even users with disabilities can have the same user experience as others. A few tips we can recommend would be integration with AI, or making text more readable, or accessible, with certain features such as text-to-speech, or speech-to-text functions.



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➤ Credible

The company and its products or services need to be trustworthy, which provides credibility for the people to want to switch over.

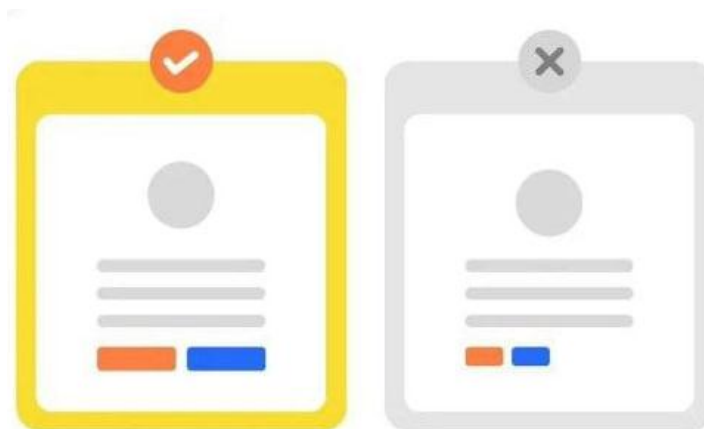
➤ *Laws of UI/UX Design:*

- *Hick's Law*



The time taken to arrive at a decision increases with the number of options. It is thus recommended to not overload the user with too many options.

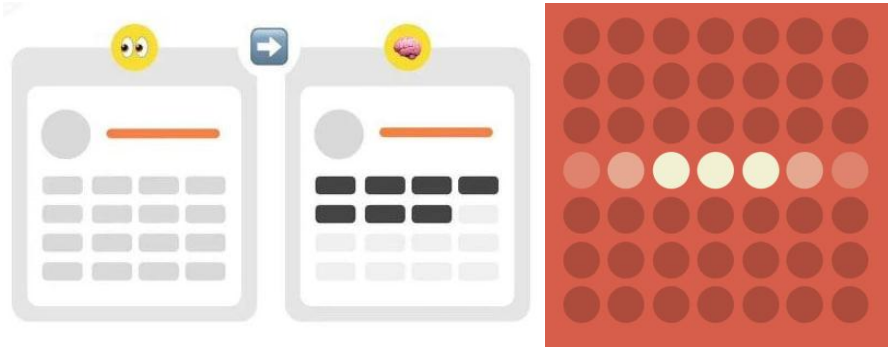
- *Fitt's Law*



The law predicts that the time required to rapidly move to a target area is a function of the ratio between the **distance to the target** and the **width of the target**.

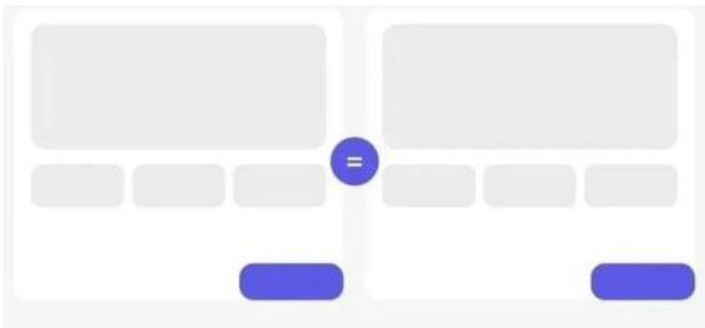
The longer the distance and the smaller the target, the longer it takes to complete the movement. Thus, it is recommended to make larger icons, while spacing them consistently and not set them too far apart.

- *Miller's Law*



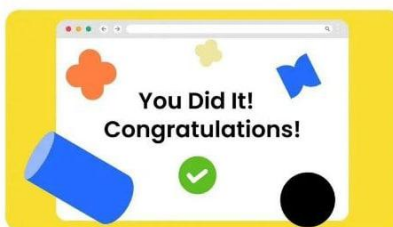
The average person can only keep 7 (plus or minus 2) items in their working memory. **Organize content into smaller chunks to help users process, understand, and memorize easily.**

- *Jakob's Law*



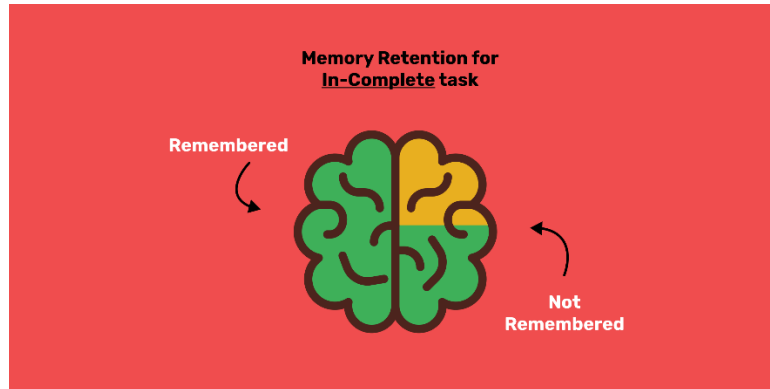
Users spend most of their time on other sites, hence they would prefer familiarity, and transfer similar expectations they have built around a familiar product. You can utilize this when trying to design and prototype your solution.

Peak-End Rule



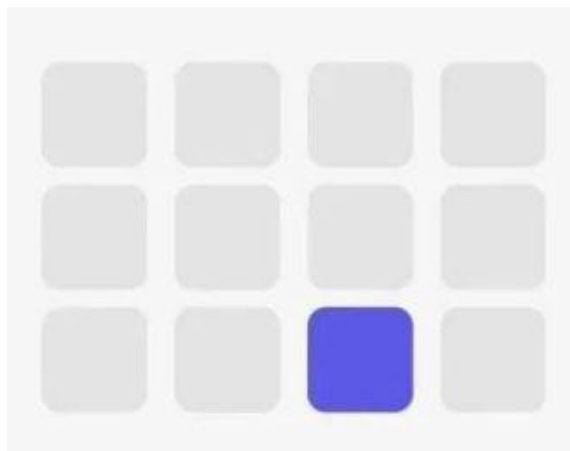
People often judge an experience, or remember it, mainly at its peak, and end, not the average or total sum of the overall experience.

Zeigarnik Effect



Zeigarnik Effect looks into how people remember information. People remember uncompleted or interrupted tasks better than completed tasks. You can consider how to tie this into your product design and prototyping.

Von Restorff Effect



Von Restorff Effect predicts that when multiple similar objects are present, the one that differs from the rest is most likely to be remembered. This ties into contrast in highlighting important information you would want the users to retain.

Other things to note that can affect UI & UX design:

Big hit applications such as Douyin or TikTok, Instagram, Telegram, and other social media platforms, or even games such as Genshin Impact that took the world by storm, all operate off a few basic principles that tie together everything covered in this document, as well as a lot more. The core idea being, minimal effort, but high engagement, leading to these applications taking advantage of a few key psychological principles and behavioral mechanisms that capitalize on how the brain reacts to **rewards, ease of use, and stimulation**.

1. Instant gratification & Dopamine Release

Dopamine Release is a core driving principle behind many addictions, such as gambling, or to social media applications, and games. Dopamine is a neurotransmitter (chemical messenger), and a complex hormone that primarily affects body, brain, and behavior. It plays a role in how we feel pleasure and rewards and ties into our unique human ability to think and plan. It helps us focus, work towards goals, and find things interesting. Behind social media applications, likes and comments, and even new content, all serve as immediate triggers for dopamine release. Since the brain craves this feeling, the instant gratification upon opening the app (new content immediately being shown on your screen), as well as the subsequent likes, comments, notifications present within, all serve to make the users want to continue engaging, as every new content, every new like or comment, offers easy and immediate dopamine release.

1.1. Social Validation & Peer Influence

The idea behind social validation ties into many other elements, such as fear of missing out (FOMO), and mainly is built upon positive reinforcement. (*Refer to operant conditioning, and social proof and gamification below*)

Instant feedback is offered through the use of social media, often in the form of likes, comments, and shares. This provides a sense of social validation, built upon positive reinforcement. This encourages users to keep engaging and posting to maintain or increase social approval. Since many of these elements and engagement metrics are made visible for everyone to see, it also fosters social interactions. Seeing others, such as friends or influencers, being constantly active on the app adds to the social pressure and creates a fear of missing out (FOMO) environment, encouraging users to open the app more, which both further contributes to the overall user experience, and results in habit formation over time.

2. Hyper-Personalized Content

A golden rule in customer service is to make the customers feel valued and feel heard. When they feel important and appreciated, they tend to stick to the same product, or a particular brand. One way that big social media platforms have achieved this is leveraging the use of algorithms and AI to deliver hyper-personalized content. By tailoring specific content, recommendations, “for you page” (FYP), to suit each individual user’s preference, it **provides relevance to the user**, while feeding them content they will **likely to be actively engaged with**. This fosters a symbiotic user relationship with the UI, and the product itself.

3. Operant Conditioning

Operant conditioning, or instrumental conditioning, is a theory of learning where behavior is influenced by its consequences. Example:

➤ Positive Reinforcement

Increases frequency of a behavior when applied:

Finish homework early == Get to movies as a reward

➤ Negative Reinforcement

Increases frequency of a behavior when a negative outcome is removed:

Alarm rings until it is silenced → Reinforcement of behavior to wake up to silence alarm

➤ Positive Punishment

Decreases frequency of a behavior when a positive outcome is added:

Teen misses curfew → Implement more rules to reduce missing curfew

➤ Negative Punishment

Decreases frequency of a behavior when a negative outcome is removed:

Abolishing the practice of clocking in and out when employees have proven they can be trusted to accurately report their time worked.

In big hit social media applications, since content has been hyper-personalised, every swipe, every click brings you to new content that the user doesn’t know about but will probably enjoy. When users don’t know when or what the next reward will be (Recall dopamine release) — this uncertainty mimics gambling, and users keep engaging with the UI, hoping for a desirable outcome (e.g., discovering a viral video or an exciting post). This variable reward

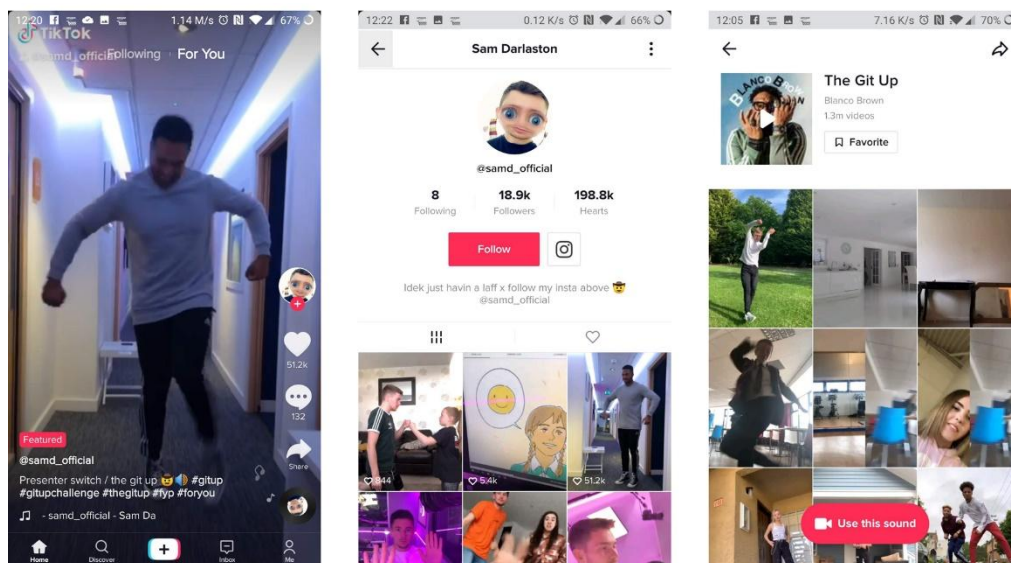
schedules are the platforms' way of integrating operant conditioning via positive reinforcement.

4. Cognitive Ease

When applications are **easy to use** and require minimal mental effort, users can engage **without feeling mentally fatigued**. Platforms such as TikTok and Instagram capitalize on this by incorporating **simplicity and familiarity** in their designs, adhering to **the 3-click rule**, **Visual Hierarchy**, **feedback & responsiveness**, **gestures and shortcuts**, **functional consistency**, **onboarding and tool tips**, **personalization features** such as dark mode (recall users in control), **aesthetic appeal**, **frictionless designs**, while including in **gamification**.

4.1. Social Proof and Gamification

Social Proof is the psychological concept that people are influenced in their decision making by the actions or behaviors of others, compelling them to act within societal norms or expectations. This phenomenon is also known as informational social influence, whereby people mimic the actions of others to try and navigate situations to achieve a desired outcome. Social media platforms leverage upon this by integrating their applications with gamification elements. Example, followers, view counts, like and dislike counts, notifications and other engagement metrics, all constitute as gamification elements where it taps into the users' interest via challenges (how to increase view counts, and like counts, etc.), achievements (*You have hit 1 million subs!*), and rewards (financial incentives, vouchers, sponsorships). All these serve to increase and retain user engagement, promote motivation via rewards (refer to instant gratification), and contribute to positive reinforcement.



Notice how TikTok also makes sure the icons are aesthetically pleasing, and intuitive!

Gamification elements often incorporate public recognition, such as leaderboards, badges, achievement systems (*You have hit 1 million subs!*), which serves to validate user efforts on both a systemic level, as well as a social level. (Refer to social validation)

5. *Habit Formation*

Habits form when new behaviors become automatic and are enacted with minimum conscious awareness (Basically it becomes second nature to the user). Applications leverage upon this and use habit-forming techniques based on the **cue-action-reward loop**:

Cue: Users receive a cue to open the app, such as a notification, or a moment of boredom (Recall dopamine release).

Action: They open the app (low effort, with only 1 tap).

Reward: They receive engaging content (FYP, reels, YouTube shorts, etc. or social validation.)

This loop strengthens with more repeated runs over time, resulting in app usage turning into a habit that becomes automatic, with users instinctively reaching for their phone and opening said applicaiton whenever the cue (e.g., boredom) occurs.

Remember, Good UI/UX Design is not something you look at and say, “Wow! That’s a great design!” Good design, when done well, **becomes invisible**. It’s only when it’s done poorly that we notice it. – Jared Spool